

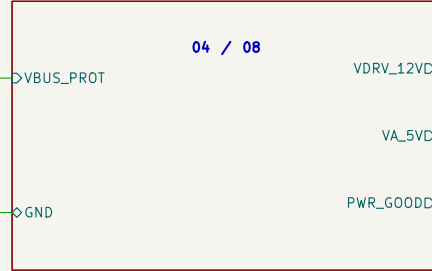
SYSTEM ARCHITECTURE – POWER FLOW / SAFETY FLOW / REAL-TIME I/O

DC Input & Protection



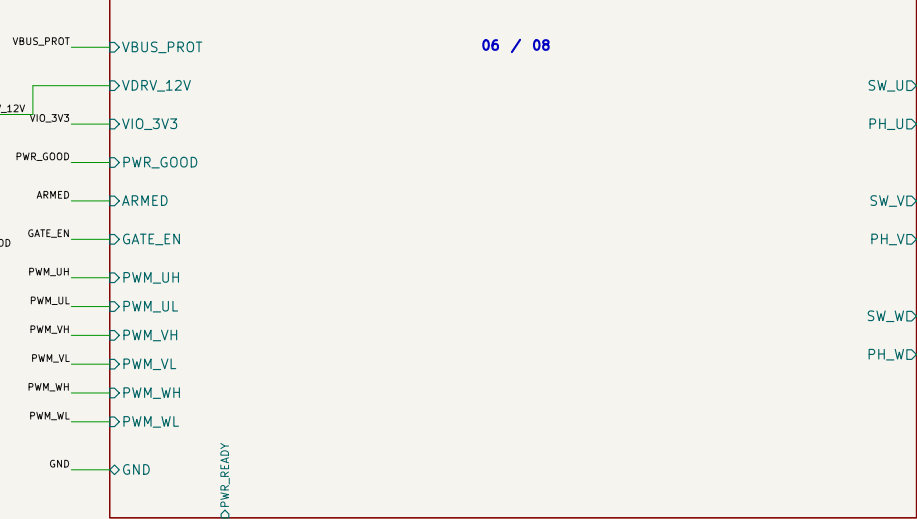
文件: power_input.kicad_sch

Auxiliary Power



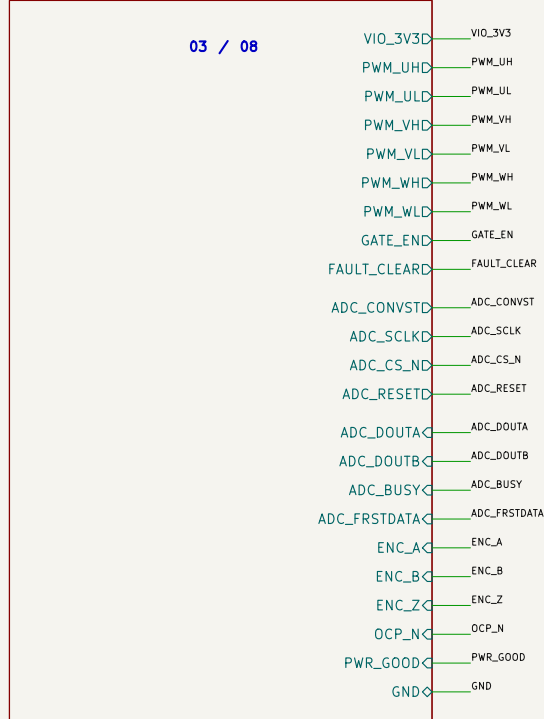
文件: aux_power.kicad_sch

Gate Driver / Inverter



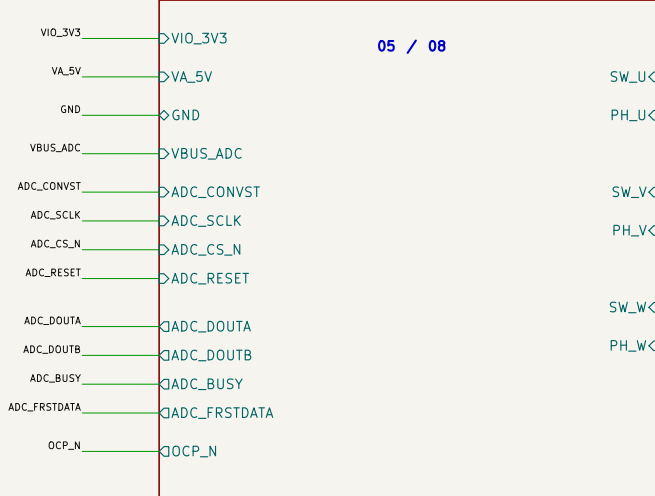
文件: gate_inverter.kicad_sch

AX7010 PL Interface

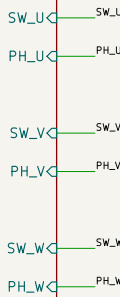


文件: ax7010_interface.kicad_sch

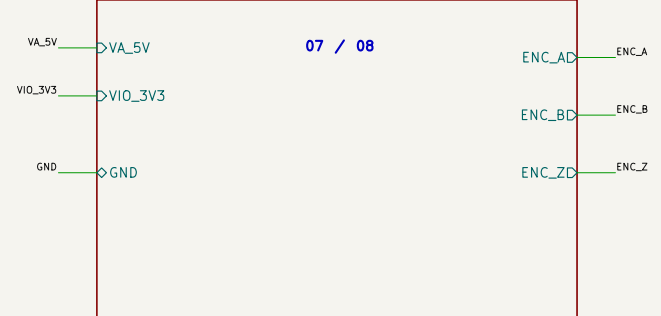
Current Sense / OCP / ADC



文件: current_adc.kicad_sch



Differential ABZ Encoder



文件: encoder.kicad_sch

48V nominal; 15..55V full-function; A4 hierarchical engineering schematic
magic-alt/fpga-servo

Sheet: /
File: ax7010_servo_reva.kicad_sch

Title: AX7010 Servo Drive Rev.A1 – System

Size: A4 Date: 2026-09-29

KiCad E.D.A. 10.0.3

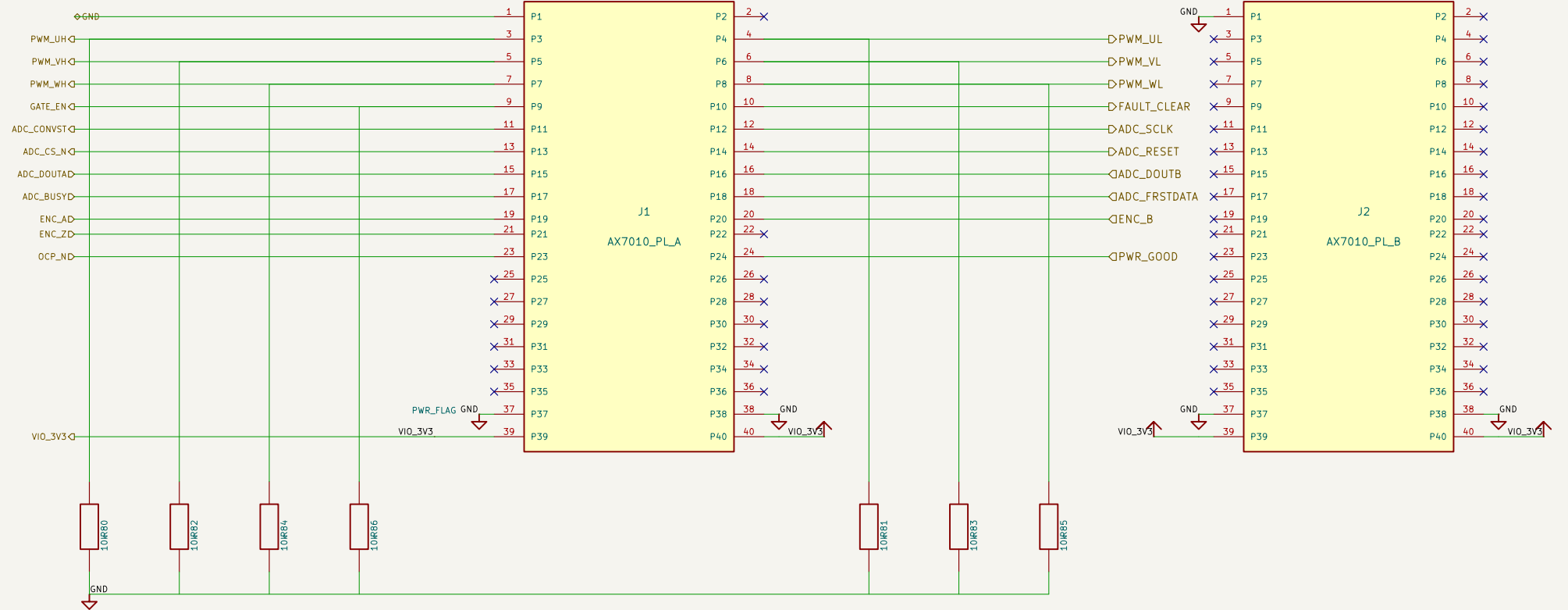
Rev: A1

Id: 1/8

ALINX AX7010 (2022): active J10 interface + physical reset-state pull-down branches

J1 = AX7010 J10

J2 = unused expansion (DNP)



R80..R86: each PWM and GATE_EN line has its own 10k pull-down.

J10 pin 2 (+5 V) remains NC; signal, power and ground pin mapping unchanged.

Two 2x20 PL expansion headers; recent AX7010 docs call these J10/J11
magic-alt/fpga-servo

Sheet: /AX7010 PL Interface/
File: ax7010_interface.kicad_sch

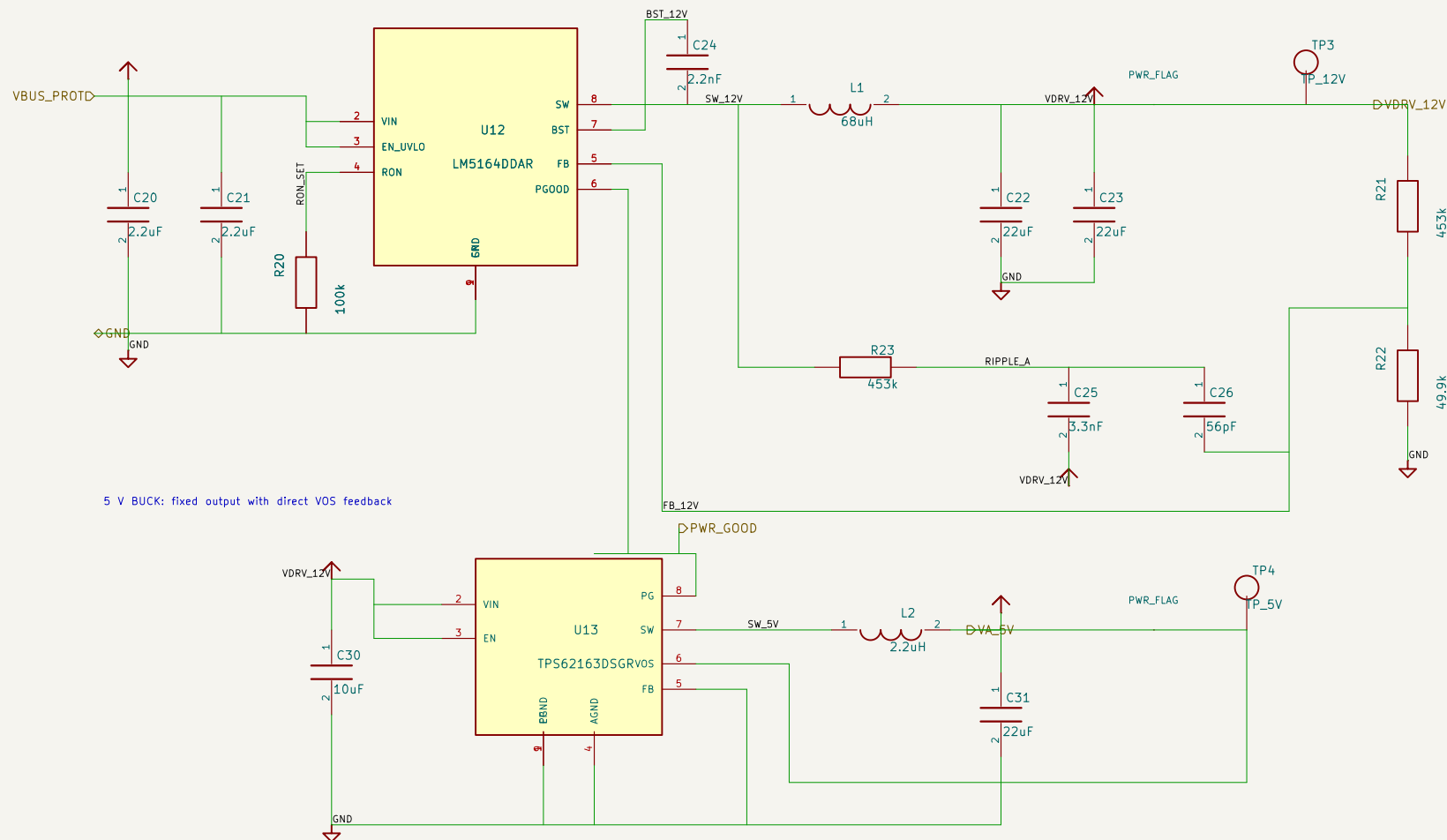
Title: Rev.A1 – AX7010 PL Interface

Size: A4 Date: 2026-09-29
KiCad E.D.A. 10.0.3

Rev: A1
Id: 3/8

VBUS_PROT -> LM5164 12V / 1A -> TPS62163 5V / 1A. PGOOD outputs are wired-AND.

12 V BUCK: input / bootstrap / output LC / feedback + ripple injection



5 V BUCK: fixed output with direct VOS feedback

LM5164 values follow TI Figure 7-1: 15..100V In, 12V/1A, 300kHz, 68uH, 2x2.2uF CIN, 2x22uF COUT, 453k/49.9k feedback.
 TPS62163 is the fixed 5V member of TPS6216x. Use 2.2uH + 22uF; connect FB to AGND and VOS to the 5V output.
 LM5164 PGOOD and TPS62163 PG share PWR_GOOD; pull up at 3.3V safety logic.
 12V input is for limited bring-up; full-function minimum is 15V.
 Hierarchy ports attach to converter nets; DNP parts remain NoConn.

LM5164 48V->12V + TPS62163 fixed 5V

magic-alt/fpga-servo

Sheet: /Auxiliary Power/

File: aux_power.kicad_sch

Title: Rev.A1 - Auxiliary Power

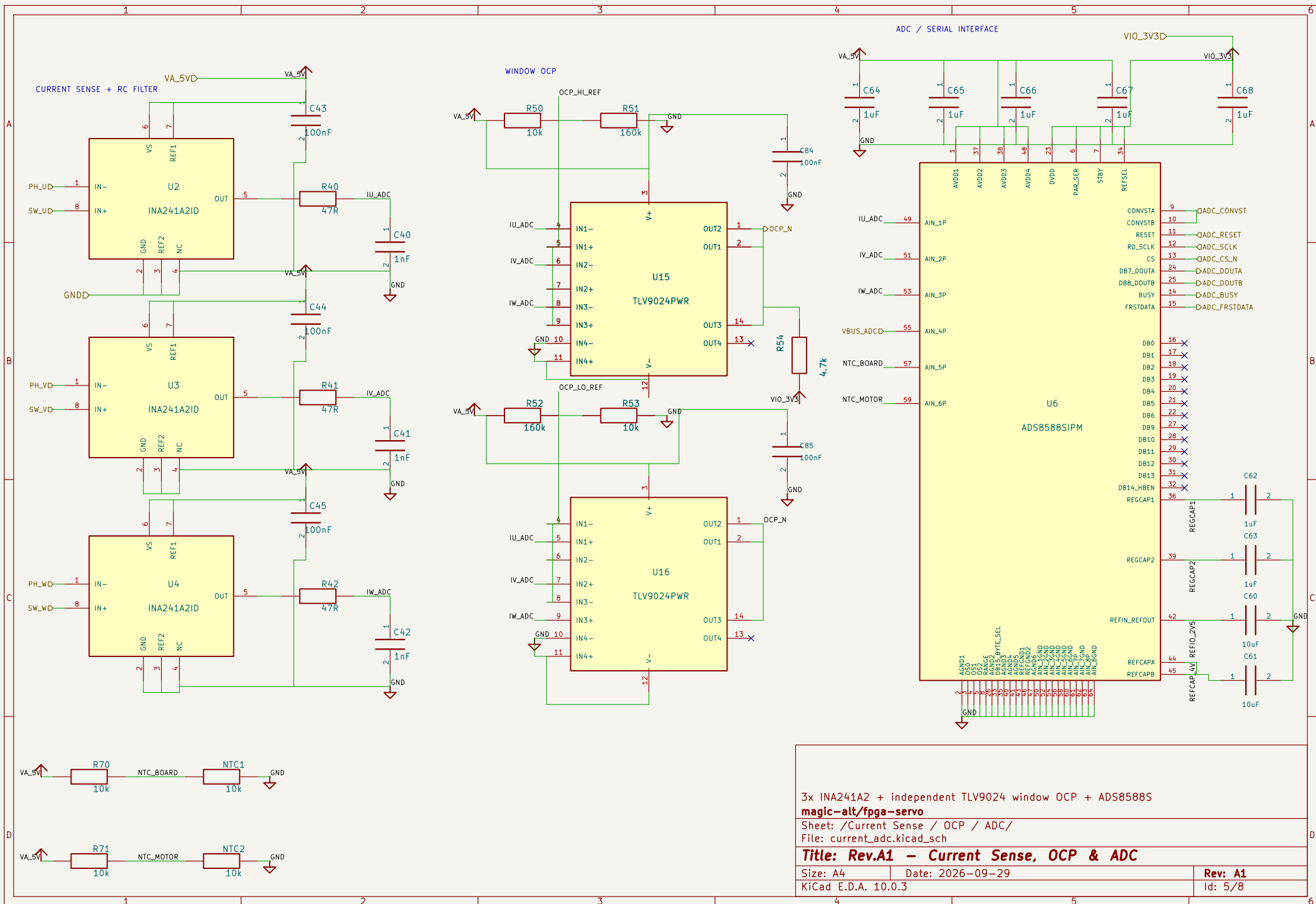
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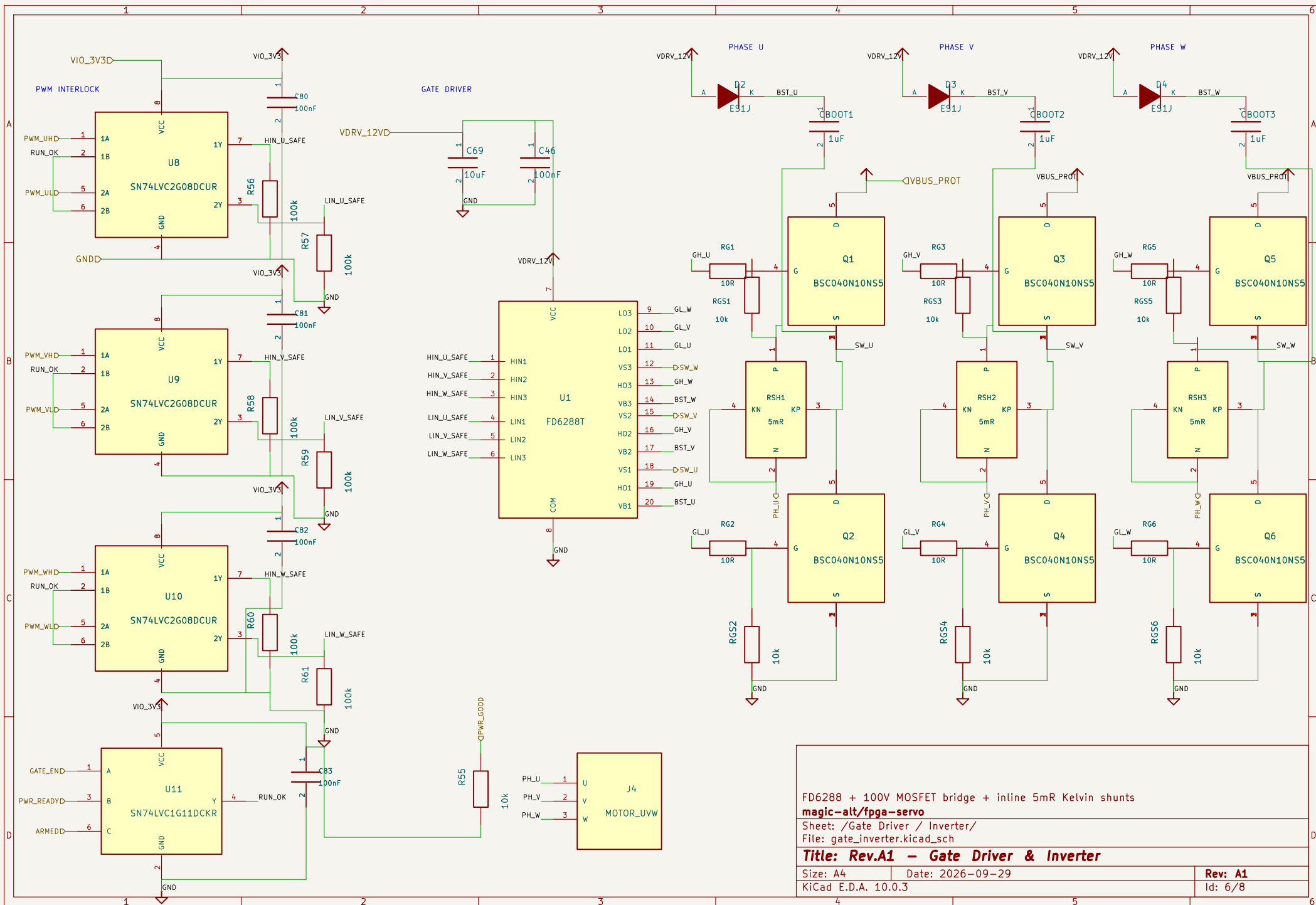
Date: 2026-09-29

KiCad E.D.A. 10.0.3

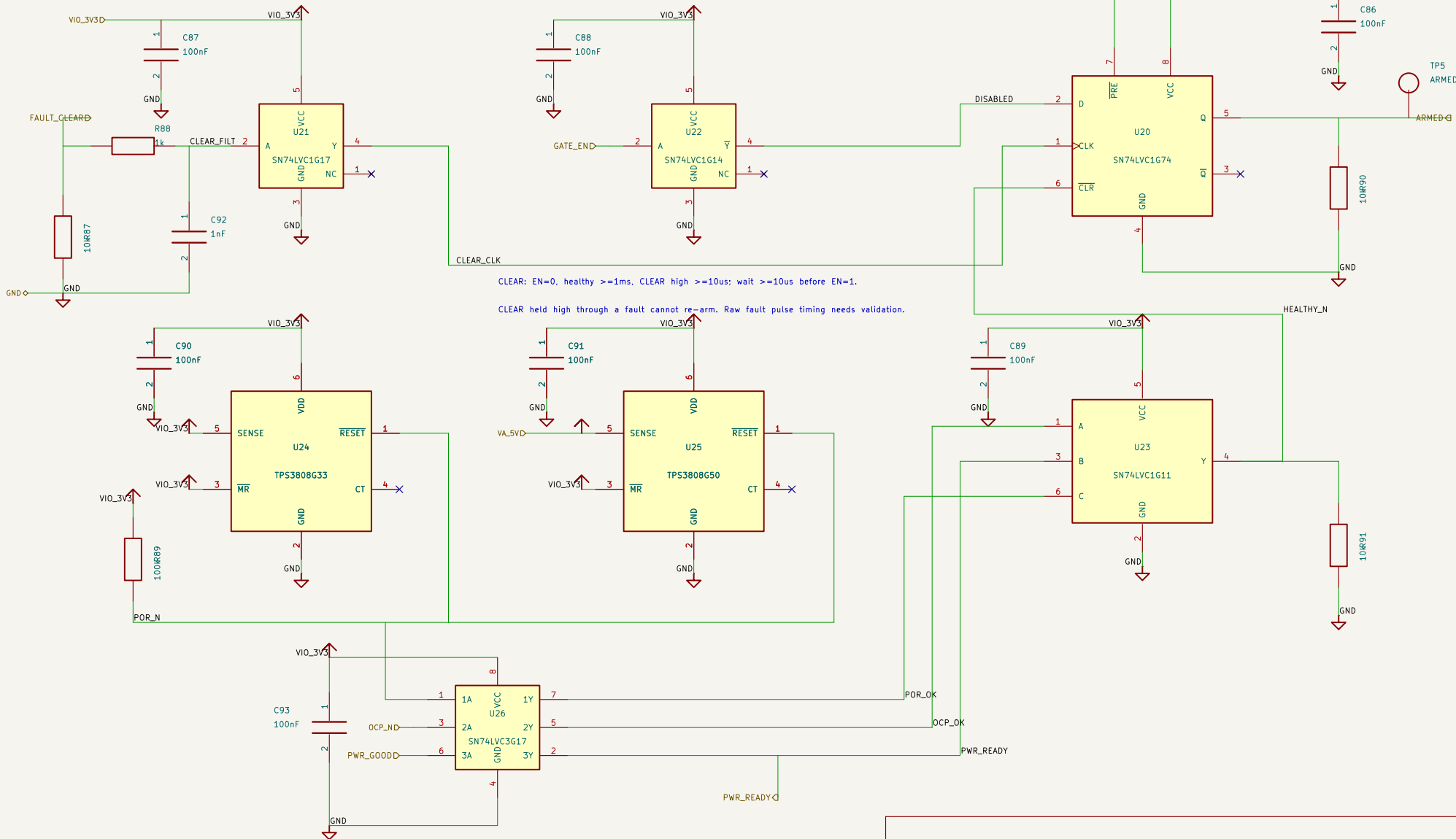
Rev: A1

Id: 4/8





HARDWARE ARM: any OCP / power fault clears Q. Recovery alone never re-arms.



CT open: reset release 12..28ms. Both supervisors use VIO; U25 senses VA5.

U20..U26: local 100nF bypass; reset release passes through U26 Schmitt buffers.

magic-alt/fpga-servo

Sheet: /OCP hardware latch/

File: ocp_latch.kicad_sch

Title: Hardware OCP latch / supervised re-arm

Size: A4

Date: 2026-10-09

Rev: A1

KiCad E.D.A. 10.0.3

Id: 8/8